

Eco-friendly covalently connected silicone hydrogel with negatively charged surface for marine anti-biofouling

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ABSTRACT

In this study, a silicone hydrogel (Gel-TA-Si-x) with amphiphilic and negatively charged surfaces was prepared by combining low surface free energy silicone with highly hydrophilic hydrogel materials using the method of one-step solvent-free bulk polymerization. The hydrogel films exhibited good mechanical properties. This was primarily reflected in the high tensile strength (1 MPa) and elongation at break (358.1 %). Notably, the film exhibited increased hydrophobicity, higher water contact angle and gradual decline in water absorption rate as the mPDMS monomer dosage increased. Despite this, the surface free energy remained low at 22.1 mJ m⁻², indicating effective fouling release capability (removal strength against pseudobarnacles was 0.16 MPa). In addition, the surface zeta potential of the hydrogel was as low as -86 mV (at pH = 10). Laboratory antifouling performance tests indicated that the surface negativity of the material enhanced its static antifouling performance and the hydrogel with a negatively charged surface exhibited high bacterial adhesion resistance of 93 % and good resistance to *Chlorella* adherence and biofilm inhibition. Favorable anti-fouling performance was observed after 180 d of marine field testing. Moreover, the covalent connection of silicone monomers with hydrogel monomers through copolymerization create a stable three-dimension crosslinking network, thereby improving the durability of materials in complex marine environments. This study provides a new idea for developing efficient and environmentally friendly silicone hydrogel antifouling materials.

1. Introduction

Marine biofouling refers to adherence and accumulation of large quantities of organisms to the surface of underwater marine facilities, causing great harm to marine equipment [1,2], such as increasing fuel consumption, accelerating material corrosion and shortening the service life of the equipment [3]. Studies have demonstrated that marine biofouling can cause tens of billions of dollars of damage annually [4]. Antifouling materials containing heavy metal ions or synergistic biocides threaten the marine environment and biological health [5]. Therefore, developing novel materials that are environmentally friendly and nontoxic and have long-lasting antifouling performance is imminent. Silicone materials are widely used as marine antifouling coatings due to their stable chemical properties in the marine environment, resistance to hydrolysis and environmentally friendly nature, which does not require biocides [6,7]. For instance, a poly (dimethylsiloxane)

(PDMS) elastomer can reduce the interaction between the dirt and the substrate due to its low surface free energy (SFE), low modulus of elasticity and low surface roughness characteristics. It can effectively prevent the sedimentation of marine organisms, which is conducive to the release of biofouling [8–10], making it an ideal marine antifouling material. However, it is only effective for high-speed ships (>15 knots), and the poor antifouling performance in static conditions limits its application [11]. Considerable efforts have been made to improve the antifouling ability of PDMS-based materials, such as introducing zwitterions [12], antimicrobial agents [13] or antifouling agents [14]. Nonetheless, the modified PDMS elastomers can easily cause serious harm to the environment owing to high amounts of antifouling substances [15,16], limiting its large-scale application.

Hydrogel is a chemically or physically cross-linked three-dimensional polymer network with excellent hydrophilicity [17,18]. It has potential applications in various fields such as biomedical materials

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[19], flexible sensors [20], electrode materials [21], tissue engineering [22], wastewater treatment [23], etc. As antifouling materials, hydrogels, such as polyethylene glycol (PEG) hydrogel [24], amphoteric hydrogel [25], nanohydrogels [26] and quaternary ammonium hydrogel [27,28] have received widespread attention in recent years. The antifouling mechanism primarily mimics the mucus layer of the biological epidermis. By forming a hydration layer on the surface of the hydrogel, it can confuse fouling organisms and prevent microorganisms from adhering [4]. However, the marine environment is complex and diverse and contains different fouling organisms, making it difficult to achieve long-term antifouling by relying only on one antifouling mechanism [29–31].

Some studies have combined silicone and hydrogel to prepare silicone-based hydrogel materials with good antifouling effects. Xue et al. [32] used a physical blending method to mix sodium polyacrylate hydrogel powder and silicone resin, which prevented the adhesion of marine organisms by forming a hydrated layer on the surface after absorbing water. Tian et al. [33] utilized silver nanoparticles as a crosslinking agent and dispersed hydrophilic polymers into a hydroxyl polydimethylsiloxane matrix to prepare silicone-based nanocomposite hydrogel hybrid material, which combines various antifouling characteristics of high hydrophilic hydrogel, functional nanoparticles and low SFE silicone to improve the antifouling performance synergistically. The antifouling test in the laboratory showed that this material killed almost 100 % of *Escherichia coli* (*E. coli*) and reduced 52 % of the settlement of *Chlorella* compared with the control sample. Additionally, this material showed excellent antifouling performance after 2 years of immersion in the South China Sea. Amphiphilic polymers can effectively reduce the adhesion of fouling organisms due to the inhomogeneous surface formed by microscopic phase separation [34]. Lu et al. [35] prepared an amphiphilic hybridized network antifouling material with epoxy silicone resin and hydrophilic hydrogel network interspersed with each other, which showed good antifouling performance both in the laboratory and in a 5-month marine field test. Since the two are usually prepared by interpenetration or physical blending, they are prone to phase separation and damage or washing away in the complex marine environment.

Qian et al. [36] reported that microorganisms first approach and adhere to the material surface by electrostatic and van der Waals forces.

Microorganisms such as bacteria are negatively charged when the pH of the external solution is greater than the isoelectric point of the microorganism. For example, *Pseudomonas fluorescens* has a zeta potential of -43.2 mV in a single extracellular polymeric substance (EPS) colloidal suspension at pH 8 [37]. *Chlorella* has a zeta potential of -30 mV in water with pH 7 [38]. Therefore, microorganisms in seawater can generally adhere quickly and tightly to positively charged surfaces due to electrostatic attraction [39]. Terada et al. [40] found that the adhesion density of *E. coli* on modified negatively charged polyethylene (PE) surfaces was 23 times lower than that on positively charged PE surfaces. They also found that the biofilms on the positively charged surfaces were dense, homogeneous and monolithic in structure, whereas the biofilm on the negatively charged surfaces was dilute, diversified and mushroom-like in structure. Therefore, negatively charged surfaces based on mutual repulsive interactions can be constructed to inhibit microbial adhesion or biofilm formation.

In this study, we prepared an amphiphilic low SFE silicone hydrogel with a negatively charged surface using the bulk polymerization method (Fig. 1(a–b)). Briefly, monomethacryloyloxypropyl terminated polydimethylsiloxane (mPDMS), N-vinyl pyrrolidone (NVP), polyethylene glycol methacrylate (PEGMA), butyl acrylate (BA), 2-phenoxyethyl acrylate (PEA) and γ -(methacryloyloxy) propyltrimethoxysilane (KH570) were firstly mixed homogeneously at room temperature, and then the polymer was obtained by heat initiation reaction. Next, the polymer was immersed in L-tartaric acid (TA) aqueous solution for surface acidification to obtain electronegative silicone hydrogel film. Among them, mPDMS provides low SFE properties. As hydrophilic components, NVP and PEGMA chain segments provided high hydrophilicity the polymer network, and the five-membered heterocyclic ring in the NVP molecule endowed it with excellent heat-resistant properties [41] and chemical stability [42]. The hydrophobic interaction of the benzene ring in the PEA molecule provided rigidity to the polymer network and improved the mechanical properties of the hydrogel [43]. The BA monomer enhanced the compatibility of the components [44] and promoted film formation [45]. More importantly, the mPDMS monomer was covalently connected with hydrogel monomers to create a durable and robust chemical crosslinking network that exhibited resistance to external forces within the intricate marine environment. Therefore, this designed a hydrogel material with good antifouling and

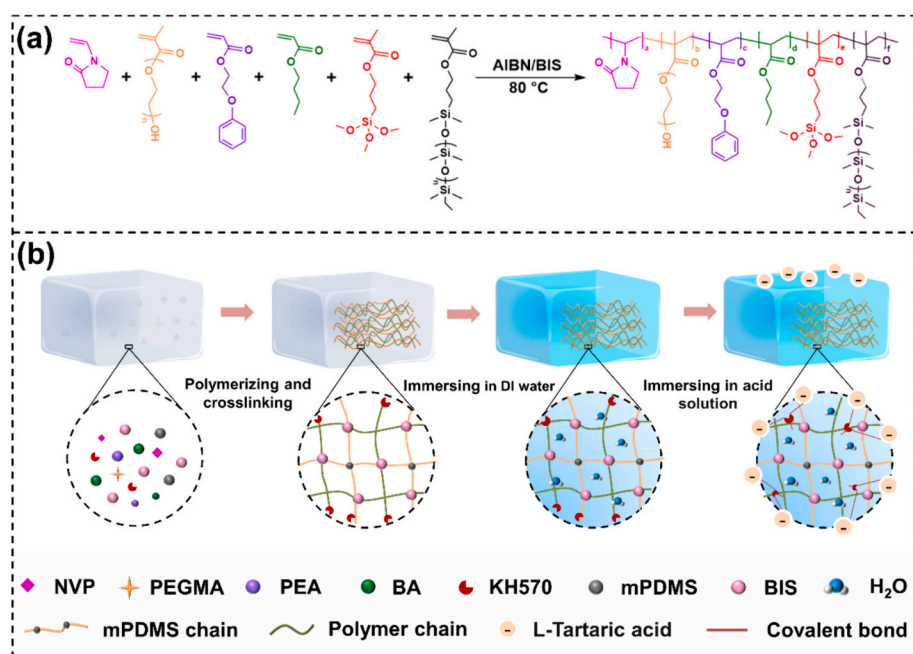


Fig. 1. Synthesis route of Gel-TA-Si-x.

mechanical properties based on a combination of green antifouling strategies.

2. Experimental section

2.1. Materials

Polyethylene glycol methacrylate (PEGMA, $M_n = \sim 360 \text{ g mol}^{-1}$), 2-phenylethyl acrylate (PEA, 98 %), monomethacryloyloxypropyl terminated polydimethylsiloxane (mPDMS, $M_n = 1000\text{--}1200 \text{ g mol}^{-1}$), γ -(methacryloyloxy) propyltrimethoxysilane (KH570, 97 %), *L*-tartaric acid (TA, 99 %) and azobisisobutyronitrile (AIBN, recrystallized after use) were purchased from Macklin Inc. (Macklin, Shanghai, China). Butyl acrylate (BA, 99 %), *N*-vinylpyrrolidone (NVP, 99 %), *N,N*-methylenebisacrylamide (BIS, 99 %) and bovine serum albumin (BSA, 98 %) were purchased from Aladdin Biochemistry and Technology Co, Ltd. (Shanghai, China). Sylgard 184 PDMS was purchased from Dow Corning. Artificial seawater (ASW) was prepared according to ASTM D1141. All other reagents were used as received from the manufacturer unless otherwise noted.

2.2. Preparation of hydrogels

BA (40 mol, 5.1 g), NVP (35 mmol, 3.9 g), PEGMA (2.5 mmol, 0.9 g), PEA (7 mmol, 1.23 g), KH570 (3 mmol, 0.75 g) and mPDMS (1 mmol, 1 g) were firstly mixed homogeneously at room temperature, and then the free radical initiator AIBN was added sequentially (0.2 wt%, 0.025 g). A cross-linking agent BIS (0.2 wt%, 0.022 g) was added and the mixture was stirred continuously for 1 h. The well-mixed solution was then poured into a polytetrafluoroethylene mold ($10 \times 10 \times 0.5 \text{ mm}$) to allow the reaction to continue for 12 h at 80°C vacuum, and the resulting silicone hydrogel films were subsequently immersed in deionized water for 24 h to remove the unreacted monomers and dissolved to equilibrium.

The obtained silicone hydrogel was immersed in a 0.01 g mL^{-1} TA solution with magnet stirring for 12 h. During the immersion process, the silane coupling agent KH570 on the hydrogel hydrolyzed to produce a hydroxyl group that underwent a condensation reaction with hydroxyl groups on the TA, which resulted in the grafting of carboxylic acid groups into a polymer network, to obtain electronegative silicone hydrogel films. The prepared hydrogels were designated as Gel-TA-Si- x ($x = 1, 2$ and 3), where 1, 2 and 3 refer to the molar ratios of mPDMS. In addition, Gel-TA without mPDMS monomer and Gel-Si-3 without immersion in TA solution were prepared for comparison.

2.3. Characterization

The attenuated total reflection-Fourier transform infrared spectroscopy (ATR-FTIR, Bruker Verter 70 spectrometer) was adopted to detect the functional groups of the hydrogel surface. Each spectrum was based on 32 scans between 4000 and 400 cm^{-1} . A zeta potential analyzer (Anton Paar surpass Austria) was used to measure the surface charge of hydrogels at a pH of 6–10. The hydrogel strip is secured in the measuring cell and a selected concentration of KCl solution was pushed under a pressure gradient. The KCl test solution drove the charge on the sample surface to the electrodes on either side, which in turn output the zeta potential from the SurPASS. Thermogravimetric analysis (TGA) was performed on an SDT-Q600 system (TA Instrument). Each sample was heated from room temperature to 800°C at a rate of $10^\circ\text{C min}^{-1}$ under a nitrogen atmosphere. Dynamic mechanical analysis (DMA) was performed on a Q800 (TA Instrument) with a thin film tension clamp from -80°C to 100°C at a heating rate of 3°C min^{-1} .

2.3.1. Water contents of hydrogels

The water content of the silicone hydrogels was determined by gravimetric analysis. First, the hydrogels were completely solubilized in

deionized water and then dried under vacuum at 40°C until constant weight. The water content (W_c) was calculated using the following equation: Eq. (1)

$$W_c = \frac{m_1 - m_0}{m_0} \times 100\% \quad (1)$$

where m_1 and m_0 are the weights of the swelled hydrogel and the dried hydrogel, respectively.

2.3.2. Tensile test

The tensile test was performed using a universal testing machine (Model AI-7000S, GOTECH, China). The samples in the tensile test were dumbbell-shaped ($75 \times 12.5 \times 4 \text{ mm}$) and the tensile rate was 50 mm min^{-1} .

2.3.3. Water contact angles and SFE

The static contact angle and SFE were calculated a contact angle goniometer (OCA 25, Dataphysics, Germany). Specifically, $3 \mu\text{L}$ of water or diiodomethane droplets were added to hydrogel at 25°C and 52 % relative humidity using the sessile method, and five measurements were performed on different regions on each sample and averaged. The SFE was calculated using the Owens-Wendt-Rabel-Kaelble method [46].

2.3.4. Scanning electron microscope (SEM) measurements

The surface morphology of samples was examined using a scanning electron microscope (SEM), preceded by gold coating under vacuum. Energy-dispersive X-ray spectroscopy (EDS) maps were recorded on the acquired SEM images.

2.3.5. Atomic force microscopy (AFM) measurements

AFM images of the samples were acquired by atomic force microscopy (Park Systems, model XE-100) using a non-contact mode in the air. Topographic images were determined based on three different regions of $10 \times 10 \mu\text{m}^2$ scanning area for each sample. NanoScope Analysis 3.0 software was used to process and analyze the images. The R_q value (the root mean square of the surface roughness height) was used to quantify the surface roughness of the hydrogels.

2.3.6. Pseudobarnacle removal test

Testing was performed based on ASTM D5618 (2011). An aluminum stud (10 mm in diameter and 10 mm in height) was glued to the hydrogel surface using a two-component epoxy adhesive. The adhesive was cured for 7 d at room temperature. The maximal shear force was measured by a force gauge (SUNDOO SH-500). Five pseudobarnacles were tested for each sample and then averaged.

2.3.7. Adhesion test

The adhesion of the hydrogel on the glass fibre-reinforced epoxy (GFE) and Q235 steel were measured using an automated tester (PosiTest AT-A, DeFelsko) according to ASTM D4541. Hydrogel films were attached to the different substrates using an underwater bonding sealant (Kraft K946). The aluminum disc (20 mm diameter) was then affixed to the hydrogel surface with a sealant. The adhesion strength data was determined by detaching them from the substrate, and the pull rate was at a speed of 0.2 MPa s^{-1} . Five different regions were tested on each sample to obtain an average value.

2.3.8. Anti-protein adsorption test

Static protein adsorption tests were performed on Gel-TA-Si- x and Gel-Si-3 using bovine serum albumin (BSA). The samples were placed in 24-well plates and irradiated under UV light for 30 min. Subsequently, 0.5 mL of 0.5 mg mL^{-1} freshly prepared BSA solution was added and immersed at 37°C for 24 h. After immersion, the samples were rinsed gently three times with PBS buffer solution to remove unabsorbed protein. The adsorbed mass of protein per unit area of the

coating ($\mu\text{g cm}^{-2}$) was calculated based on the decreased concentration measured at 280 nm on a UV-vis spectrometer (Jinghua UV1800PC, Shanghai, China). Each value was the average of at least five measurements on different samples. The corresponding BSA adsorption capacity was calculated using Eq. (2):

$$Q_{BSA} = \frac{V_{BSA} \times (C_0 - C_1)}{S_{BSA}} \quad (2)$$

where Q_{BSA} ($\mu\text{g cm}^{-2}$) and V_{BSA} (L) referred to the adsorption capacity of BSA and the volume of BSA solution. S_{BSA} (cm^2) represented the membrane area. The concentrations of BSA before and after static adsorption were expressed as C_0 (g L^{-1}) and C_1 (g L^{-1}), respectively.

2.3.9. Antibacterial assays

Pseudomonas aeruginosa (P. sp.), *E. coli* and *Staphylococcus aureus* (S. aureus) were selected as representative bacteria. They were incubated in Luria-Bertani (LB) medium at 37 °C for 10 h, and the concentration of bacteria in the medium was expressed as colony-forming units (cfu). Subsequently, the bacterial bodies were separated from the medium by centrifugation at 1324 $\times g$ for 5 min. The bacteria were then diluted and suspended in 0.1 mol L^{-1} phosphate-buffer saline (PBS) (8.0 g L^{-1} NaCl, 0.2 g L^{-1} KCl, 1.44 g L^{-1} Na_2HPO_4 and 0.44 g L^{-1} KH_2PO_4 in distilled water). Finally, PBS buffer containing 10^7 cfu mL^{-1} of bacteria was used for antibacterial experiments. Bare glass slides (as a blank) and slides coated with PDMS elastomer (as a reference), and hydrogels were first sterilized under ultraviolet (UV) light for 30 min, and then immersed in prepared bacterial suspension at 25 °C for 6 h. Samples were stained with the LIVE/DEAD BacLight Bacterial Viability Kit for 15 min, and rinsed twice with PBS buffer to wash off the weakly adherent bacteria and excess stain. Finally, the attached bacterial cells were observed under a fluorescence microscope (BX-51, Olympus, Japan) and five fields of view were selected on parallel samples for counting respectively. The anti-adhesion rates (R) of the bacteria were calculated using the following equation: Eq. (3):

$$R = \frac{N_0 - N_1}{N_0} \times 100\% \quad (3)$$

where N_0 and N_1 are the number of bacteria on glass slides and hydrogels, respectively. All media were sterilized at 121 °C for 30 min before use. An AIRTECH clean bench was used after UV irradiation for 30 min during the inoculation and assembly steps.

2.3.10. Anti-biofilm test

The anti-biofilm properties of hydrogels were tested using P. sp. bacteria. Each hydrogel was placed in a 24-well plate and incubated in 0.5 mL bacterial suspension (10^8 cfu mL^{-1}) at 37 °C for 24 h. Next, biofilms were rinsed thrice with PBS to remove the suspended bacteria, stained with 0.1 % crystal violet solution for 15 min and rinsed thrice with PBS solution. The stained biofilms were observed under the fluorescence microscope (Olympus BX-51, Japan). Finally, the crystal violet stain was extracted with 2 mL absolute ethanol and measured using a UV-vis spectrometer (Jinghua UV1800PC, Shanghai, China) at 590 nm to obtain the optical density (OD) value.

2.3.11. Anti-chlorella test

Chlorella was cultured in F/2 medium at 23 °C in a 12:12 h light/dark cycle. Bare glass slides (as a blank), slides coated with PDMS elastomer (as a reference) and prepared samples were placed in *Chlorella* suspension (1.05×10^7 cells mL^{-1}) and incubated for 24 h. After 24 h. The incubated samples were gently washed with ASW to remove unattached algae and the adherent algae was examined under a fluorescence microscope (Olympus BX-51, Japan). The counts were prepared for the five fields of view on each of the three parallel samples and the average density of the attached algae was recorded.

2.3.12. Marine field test

Samples were adhered to epoxy resin plates using a Kraft (K946) underwater adhesive sealant, and bare epoxy resin plates and epoxy resin plates coated with PDMS elastomers were used as controls. The marine field test was conducted in Qingdao Port, China (24°33'N, 118°09'E). Epoxy resin panels were immersed in 1 m deep seawater for 6 months (from January to July) for attachment of fouling organisms. Epoxy resin plates were then periodically removed from the sea and photographed using a digital camera to observe the deposition of biofouling organisms on the plates. In addition, panels were cleaned with a high-pressure water gun at 120 and 180 d.

3. Results and discussion

3.1. Structural characterization

ATR-FTIR was used to characterize the chemical structure of Gel-TA-Si-x (Fig. 2a). The O—H stretching vibration peaks were observed at 3420 cm^{-1} . Absorption bands at 2960–2880 cm^{-1} were attributed to the stretching vibration of C—H. The characteristic absorption peaks at 1726 cm^{-1} and 1662 cm^{-1} represented the stretching vibrations of ester carbonyl and ketone carbonyl groups and the peak at 1163 cm^{-1} was related to the stretching vibration of the ester C—O group. The absorption peak at 1257 cm^{-1} was associated with the stretching vibration of the C—N group. The absorption peak at 1460 cm^{-1} was related to the C—C backbone on the benzene ring. Gel-Si-1 sample was prepared without immersion in water for comparison. The characteristic absorption peak at 1190 cm^{-1} was ascribed to the stretching vibration of Si—O—C and the intensity of the absorption peak at 1190 cm^{-1} in the Gel-Si-1 sample was significantly higher than Gel-TA-Si-1 (Fig. S1). This indicates that the silane moiety in KH570 was hydrolyzed, leading to the formation of reduced Si—O—C absorption peaks. The hydrolyzed siloxane condensed with the TA molecule, successfully grafting carboxylic acid groups onto the surface, as evidenced by the intensified C=O absorption peak at 1662 cm^{-1} . Absorption bands at 1020 and 794 cm^{-1} were attributed to the stretching vibration of Si—O—Si and Si—C. Absorption peaks at 1020 cm^{-1} and 794 cm^{-1} were attributed to the stretching

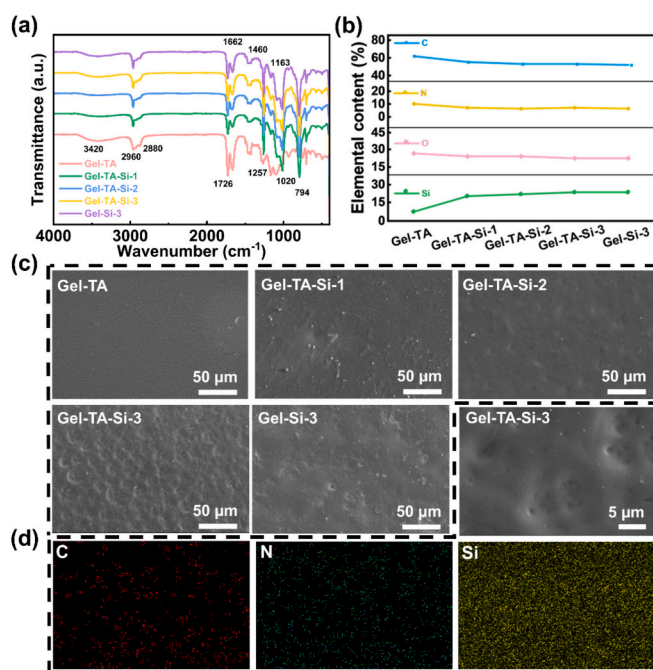


Fig. 2. Characterization and morphology of silicone hydrogel (a) ATR-FTIR, (b) elemental content, (c) SEM of Gel-TA-Si-x and Gel-Si-3, and (d) EDS elemental mapping of Gel-TA-Si-3.

vibration of Gel-TA-Si-1, Gel-TA-Si-2, Gel-TA-Si-3 and Gel-Si-3, however, Gel-TA had no characteristic peaks, indicating successful introduction of silicon monomers. ATR-FTIR demonstrated the successful synthesis of Gel-TA-Si-x polymers.

SEM images revealed that the surface morphology of the Gel-TA was relatively flat. Microscopic phase separation occurred on the surfaces of Gel-TA-Si-1, Gel-TA-Si-2, Gel-TA-Si-3 and Gel-Si-3 (Fig. 2c). The hydrogel was tested by EDS mapping to further prove the polymer generation. It was found that C, N, and Si elements were uniformly distributed on the surface of the samples (Figs. 2d and S2), confirming good compatibility of the components. The content of the Si element increased with an increase in the amount of silicon monomer (Fig. 2b), consistent with the results of the feeding ratio of the mPDMS monomer.

3.2. Mechanical properties

Subsequently, the mechanical properties of different hydrogels were tested to ensure that external forces did not damage the hydrogel during the application process. As shown in Fig. 3a–c, the elongation and toughness of Gel-TA without mPDMS were 76.3 % and 0.29 MJ m⁻³ respectively, and the tensile strength was lower. The tensile strength of Gel-TA-Si-1 was 1.0 MPa. Gel-TA-Si-3 had the highest elongation (358.1 %) and the largest toughness (1.41 MJ m⁻³), perhaps due to the good flexibility of the Si-O-Si chain segments in mPDMS, resulting in high elongation and toughness of the material. The elongation and toughness of the hydrogels increased with an increase in the amount of mPDMS. The elongation increased from 197.4 % to 358.1 % and the toughness from 1.13 MJ m⁻³ to 1.41 MJ m⁻³ when the molar mass was increased from 1 mmol (Gel-TA-Si-1) to 3 mmol (Gel-TA-Si-3). This may be attributed to the complex chain entanglement caused by longer Si-O-Si chain segments in mPDMS; the stronger the chain entanglement caused by the higher dosage of mPDMS, the better the mechanical properties [47]. Therefore, the introduction of silicone monomers can effectively improve the mechanical properties of hydrogels. The typical

mechanical property parameters of Gel-TA-Si-x, including the tensile strength, the elongation at break, the modulus of elasticity and the toughness are shown in Fig. 3d.

The thermal stability of Gel-TA-Si-x was analyzed using the TGA curve (Fig. 3e). The weight loss before 150 °C was mainly the loss of water [48]. The main mass loss occurred between 300 and 450 °C, probably caused by the breakage of the polymer backbone. The derivative thermogravimetric analysis (DTG) curve also shows two decomposition temperatures, thus, the decomposition of the polymer backbone had only one stage, indicating that the polymer has good thermal stability. The relationship between the energy storage modulus (E') and the temperature of Gel-TA-Si-x is displayed in Fig. 3f. E' decreased with increasing temperature, the higher the temperature, the higher the polymer viscosity and the higher the fluidity. In addition, E' was associated with the content of mPDMS; the higher the silicone monomer content, the higher the content of mPDMS, resulting in a lower E' . The temperature dependence of the loss tangent ($\tan\delta$) was presented in Fig. 3g. Notably, $\tan\delta$ values were consistently below 1, which demonstrated that all samples were mainly elastically rather than viscously deformed [49]. Furthermore, the peak of $\tan\delta$ corresponded to the glass transition temperature (T_g). The order of the T_g values of the samples was as follows: Gel-TA (20.75 °C) > Gel-TA-Si-1 (10.77 °C) > Gel-TA-Si-2 (-4.14 °C) > Gel-TA-Si-3 (-7.58 °C). It was found that the peak of $\tan\delta$ shifted to the left and T_g gradually decreased with increasing content of mPDMS, which may be attributed to the increased flexibility of the polymer. mPDMS contains long Si-O-Si chain segments with long bonds, large bond angles and small internal rotation barriers, indicating good conformational flexibility [50]. Additionally, longer siloxane chain segments provided better flexibility and fluidity with an increase in mPDMS content [51], ultimately resulting in enhanced flexibility of the polymer.

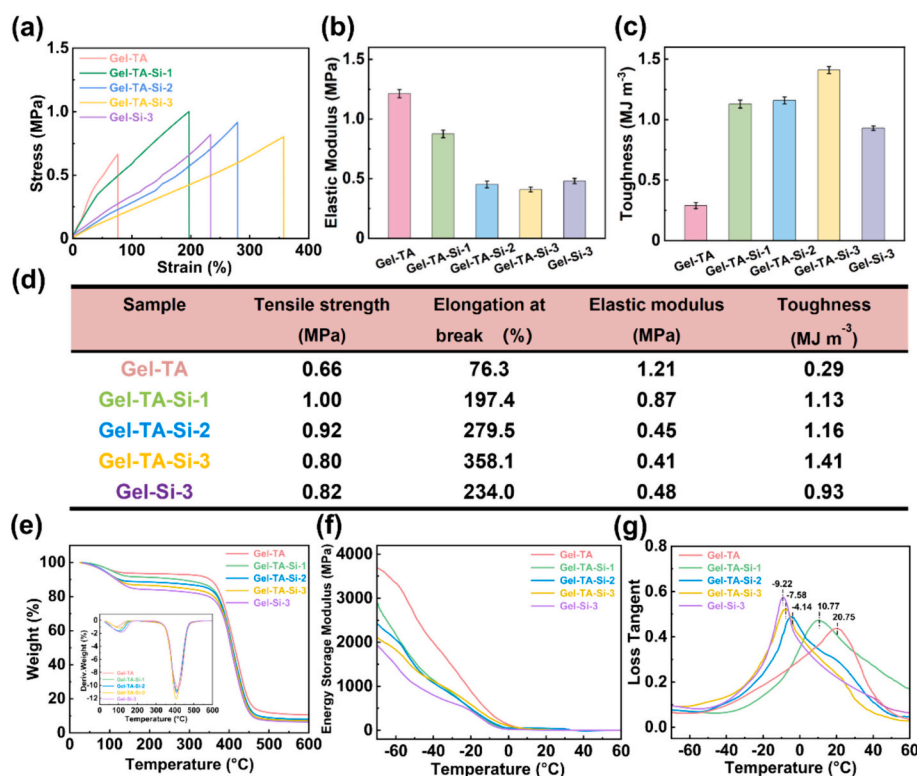


Fig. 3. (a) Stress-strain curves. (b) Modulus of elasticity. (c) Toughness of Gel-TA-Si-x and Gel-Si-3. (d) Summary of the mechanical properties of different samples. (e) TG and DTG curves of Gel-TA-Si-x and Gel-Si-3. (f) E' -temperature curves, and (g) $\tan\delta$ -temperature curves of Gel-TA-Si-x and Gel-Si-3.

3.3. Surface properties

There are complex and multifactorial interactions between fouling organisms and material surfaces. Surface properties such as wettability, surface energy and roughness of the material surface may affect adhesion of fouling organisms [52]. Therefore, several tests were conducted to evaluate the surface properties of the Gel—Si—3 and Gel-TA-Si-x. The static water contact angle (WCA) of the silicone hydrogel is shown in Fig. 4a. The WCA of the PDMS surface was $\sim 116.5^\circ$ [53]. WCA values of Gel-TA, Gel-TA-Si-1, Gel-TA-Si-2 and Gel-TA-Si-3 were $\sim 81.4^\circ$, $\sim 92.4^\circ$, $\sim 95.9^\circ$, and $\sim 100.7^\circ$ respectively, which showed that the hydrogels exhibited higher hydrophilicity than the PDMS. The WCA of the hydrogels gradually increased with an increase in mPDMS content, as the surface became more hydrophobic. This finding is consistent with the results from the water content test (the water content gradually decreased from 38.1 % to 26.5 %) (Fig. 4c). Gel-TA-Si-1 hydrogel before and after swelling was shown in Fig. S3. This may be explained by the gradual decrease in water absorption of the hydrogels, enhanced hydrophobicity of the sample surfaces and the gradual increase in the WCA. As shown in Fig. 4b the SFE values of Gel-TA, Gel-TA-Si-1, Gel-TA-Si-2 and Gel-TA-Si-3 were ~ 35.7 , ~ 31.1 , ~ 28.4 and ~ 22.1 mJ m^{-2} , respectively. This demonstrates that the introduction of mPDMS monomer remarkably reduces the SFE of the hydrogels. Table S1 showed the WCA and SFE of various material surfaces.

Surface roughness plays a crucial role in preventing the adhesion of fouling organisms, as they cannot adhere to smooth surfaces. The surface roughness of each sample was quantitatively examined with AFM (Fig. 5(a–e)). The R_q of Gel-TA was as low as 0.01 μm with a relatively smooth surface. Notably, the R_q of hydrogels increased from ~ 0.12 to ~ 0.40 μm with an increase in mPDMS content, which can be attributed to enhanced incompatibility among the components in the polymer network by the introduction of mPDMS [54]. Discontinuous bright and dark regions observed on the surfaces of Gel-TA-Si-x and Gel—Si—3 corresponded to hydrophobic mPDMS segments and hydrophilic hydrogel segments, respectively, indicating the microphase segregation

on the surface. This is also consistent with SEM result. Due to the low surface energy of mPDMS and its incompatibility with hydrogel matrices, it had a trend to migrate towards the surface instead of being buried in the bulk. The degree of microphase separation increased with higher mPDMS content, leading to self-enrichment on the surface and an associated increase in roughness.

The removal strength of pseudobarnacle, such as surface roughness, surface free energy and modulus of elasticity are affected by [55]. A previous study reported that materials with $\text{SFE} < 30$ mN/m can effectively inhibit barnacle attachment [56]. PDMS has a good fouling desorption ability, with a low removal strength of ~ 0.22 MPa. The removal strength of Gel-TA, Gel-TA-Si-1 and Gel-TA-Si-2 was significantly higher than that of PDMS (Fig. 4d), perhaps due to the increase in surface energy and roughness, which is unfavorable for the desorption of fouling organisms [57]. The surface roughness of Gel-TA-Si-3 and Gel—Si—3 were relatively higher, but the removal strengths of Gel-TA-Si-3 and Gel—Si—3 were only 0.16 and 0.17 MPa, respectively. This could be due to their low surface free energy and modulus of elasticity, which require less force to remove [58].

3.4. Adhesion properties

To examine the durability of the material in the marine environment, the hydrogel sample was tested in terms of its adhesion strength after immersion in AWS solution for 14 days. As shown in Fig. S4, Gel-TA-Si-1 exhibited the highest adhesion strength (GFE: ~ 1.01 MPa, steel: ~ 1.12 MPa). Gel-TA, Gel-TA-Si-x and Gel—Si—3 hydrogels had higher adhesion strengths compared to PDMS (GFE: ~ 0.43 MPa, steel: ~ 0.61 MPa). This result demonstrates that all hydrogels were stable in the marine environment.

3.5. Anti-biofouling properties

Proteins are a primary component of the initial marine fouling stage, nourishing bacteria and other microorganisms [2]. The BSA is a

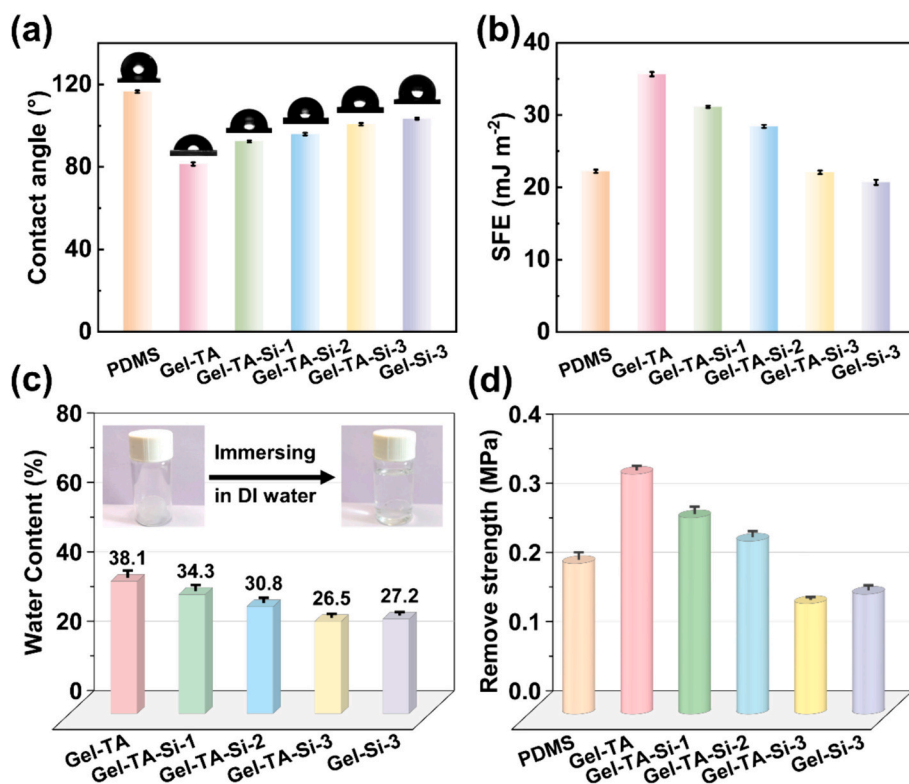


Fig. 4. Surface properties of hydrogels (a) water contact angle of Gel-TA-Si-x and Gel—Si—3, (b) SFE, (c) water content and (d) pseudobarnacles removal strength.

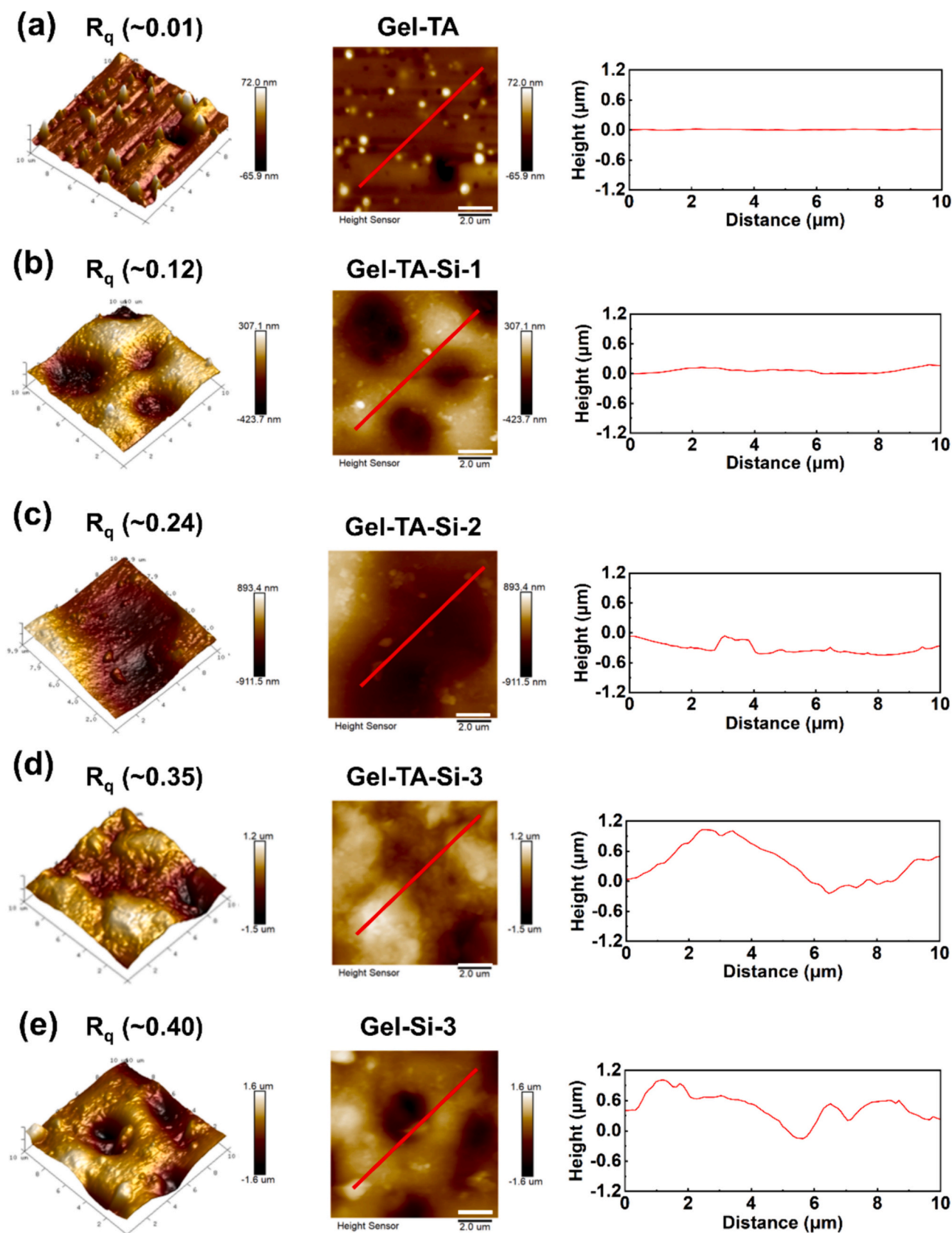


Fig. 5. AFM profiles of (a) Gel-TA, (b) Gel-TA-Si-1, (c) Gel-TA-Si-2, (d) Gel-TA-Si-3 and (e) Gel-Si-3 hydrogel evaluated in amplitude modulation mode in air (left to right: 3D topography image ($10 \times 10 \mu\text{m}^2$) and R_q (nm), 2D topography image ($10 \times 10 \mu\text{m}^2$), height profile across the red line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

commonly used protein model for anti-protein testing. To evaluate the anti-protein adsorption of silicone hydrogels, we used BSA. Fig. S5 indicates that the amount of BSA adsorbed by the PDMS was approximately $93.54 \mu\text{g cm}^{-2}$, while the amount of BSA adsorbed by Gel-TA-Si-x and Gel-Si-3 was significantly decreased. This may be due to the space repulsion forces caused by the hydrophilic PEG chain segments or the formation of a film with water molecules to resist protein adhesion [59]. The amounts of BSA adsorbed for Gel-TA, Gel-TA-Si-1, Gel-TA-Si-2, and Gel-TA-Si-3 were approximately 40.02, 12.03, 7.42, and $6.62 \mu\text{g cm}^{-2}$, respectively. This reduction can be attributed to microphase separation and changes in the surface conformation of the hydrogels in the aqueous environment [60].

P. sp., *E. coli* and *S. aureus* were selected as model bacteria because they are commonly used in the evaluation of antimicrobial antifouling materials [26,61]. The results are shown in Fig. 6(a–c). Fluorescence images reveal that a large number of bacteria *P. sp.*, *E. coli* and *S. aureus* adhered to the bare glass slide (as a blank) and PDMS (as a reference), whereas fewer bacteria adhered to the surface of Gel-TA-Si-x.

Additionally, the bacteria attached to the hydrogel surface decreased gradually with an increase in the amount of mPDMS monomer. The anti-adhesion rates of Gel-TA-Si-3 for *P. sp.*, *E. coli* and *S. aureus* were 90.5 ± 2.2 , 89.3 ± 2.3 and 93.4 ± 1.9 % (Fig. 6e). This phenomenon may be attributed to the hydrophilic polyethylene glycol (PEG) chain segments contained in PEGMA which promoted the adsorption of water molecules to form a hydrated layer, which effectively prevented bacterial adhesion [4,62]. Moreover, the low SFE and low elastic modulus properties of silicone hydrogels reduced the initial bacterial adhesion [11,63]. An

inhomogeneous dual hydrophobic and hydrophilic surface is formed on the microscopic scale due to amphiphilic polymer chain segments in the hydrogel, which effectively prevents bacterial adhesion [64]. Consequently, Gel-TA-Si-x combines multiple antifouling mechanisms that synergistically resist bacterial adhesion. It also found that the surface zeta potential of Gel-TA-Si-x was significantly lower than that of Gel-Si-3 (Fig. 6g). Given that the pH range of seawater falls within the range of 7.9 to 8.4, the potentials were examined specifically for a pH value of 8. For example, the zeta potential of Gel-Si-3, Gel-TA, Gel-TA-Si-1, Gel-TA-Si-2 and Gel-TA-Si-3 were -19 ± 2.67 , -27 ± 1.91 , -69 ± 2.55 , -37 ± 1.46 and -33 ± 2.88 mV. The zeta potential of PDMS was -57 ± 2.38 mV (at pH = 8), which is attributed to PDMS contain contained a small amount of fumed silica (Si-OH) and these substances could be ionized in solution to form charged SiO^- sites on the surface [65]. In contrast, the zeta potential of Gel-TA-Si-1 was -69 ± 2.55 mV (at pH = 8), which had a more negative surface charge. Since the majority of bacteria are anionic in nature [66], they experience mutual repulsion from silicone hydrogel surfaces, which also carry a negative charge. Huang et al. [67] found that the antifouling performance of the film was increased owing to the enhancement of its negative charge and the repulsive force generated by the negatively charged fouling. Therefore, Gel-TA-Si-1 showed better antimicrobial effects than PDMS. The antibacterial effect of the acidified hydrogel was superior to that of the unacidified hydrogel, which aligns with the results of the antibacterial rate test. The anti-adhesion rates of Gel-TA-Si-3 for *P. sp.*, *E. coli* and *S. aureus* were higher than Gel-Si-3 (53.2 ± 2.1 , 58.5 ± 3.7 , and 54.9 ± 3.2 %).

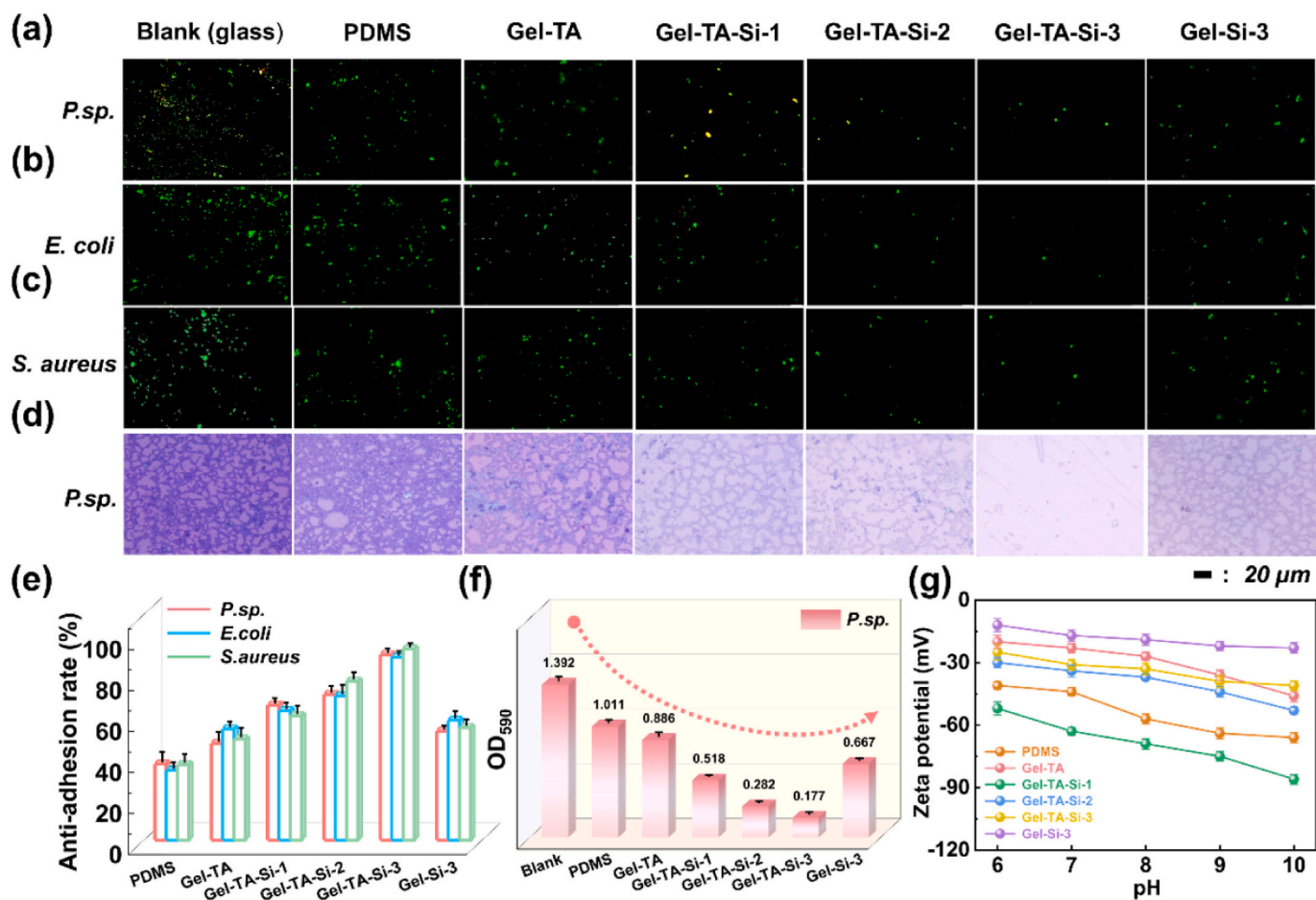


Fig. 6. Fluorescence images of (a) *P. sp.*, (b) *E. coli* and (c) *S. aureus* adhered to each hydrogel surface. (d) *P. sp.* biofilm grown on each hydrogel surface stained with crystal violet. The scale bar is 20 μm . (e) Anti-bacterial adherence rate. (f) OD₅₉₀ values against *P. sp.* of each hydrogel surface. (g) Surface zeta potentials of PDMS, Gel-TA-Si-x and Gel-Si-3. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In the early stages of biofouling, bacteria accumulate on the target surface to form biofilms. Therefore, anti-biofilm tests were performed to evaluate the performance of Gel-TA-Si-x in inhibiting bacterial biofilm growth. As shown in Fig. 6(d), the surfaces of bare glass slides, PDMS and Gel-TA stained with crystal violet showed a dark purple color and exhibited high OD₅₉₀ values of approximately 1.392, 1.011 and 0.886, respectively (Fig. 6(f)). This indicated that a large number of biofilms were formed on their surfaces. However, Gel-TA-Si-1, Gel-TA-Si-2 and Gel-TA-Si-3 gradually became lighter in color, and their OD₅₉₀ values gradually decreased from 0.518 (Gel-Si-1) to 0.177 (Gel-TA-Si-3), suggesting that Gel-TA-Si-x disrupted the formed biofilm. Considering that the hydrogel contained hydrophobic and hydrophilic units, microphase separation occurred at the microscopic scale. This inhomogeneous surface effectively prevented biofilm adhesion [68]. Furthermore, Gel-Si-3 had an OD₅₉₀ value of 0.667, which was remarkably higher than that of Gel-TA-Si-3. This is attributed to the fact that the surfaces of bacterial biofilms are often negatively charged [69], making them repel each other and hydrogels with negatively charged surfaces, which enhances their resistance to biofilms. Therefore, the amphiphilic, negatively charged, low SFE and smooth surface properties of silicone hydrogel synergistically resist bacterial adhesion and biofilm formation.

3.6. Anti-chlorella properties

The hydrogel was effective in reducing the attachment of *Chlorella* [70]. Notably, Fig. 7(a-b) showed that the settlements of *Chlorella* on the surface of Gel-TA-Si-1 (1 d: ~51 cells mm⁻², 7 d: ~42 cells mm⁻², 14 d: ~63 cells mm⁻², 28 d: ~72 cells mm⁻²), Gel-TA-Si-2 (1 d: ~29 cells mm⁻², 7 d: ~34 cells mm⁻², 14 d: ~25 cells mm⁻², 28 d: ~37 cells mm⁻²) and Gel-TA-Si-3 (1 d: ~8 cells mm⁻², 7 d: ~15 cells mm⁻², 14 d: ~9 cells mm⁻², 28 d: ~11 cells mm⁻²) gradually decreased with an increase in mPDMS dosage, which could be attributed to the steady decrease in SFE and elastic modulus of Gel-TA-Si-x [30,71]. In addition, the coverage areas of *Chlorella* were significantly reduced on the surface of Gel-TA-Si-3 compared with Gel-Si-3. This reduction may be because *Chlorella* is negatively charged in seawater-like environments at pH = 8 and repels hydrogels with negatively charged, thus effectively reducing the attachment of *Chlorella*. Overall, these findings indicate that Gel-TA-Si-3 had better antimicrobial, anti-chlorella and anti-biofilm properties.

3.7. Marine field experiments

A marine field test was conducted in Qingdao Port, China (24°33'N, 118°09'E) to further confirm the static antifouling performance of the

silicone hydrogel. Epoxy resin panels were immersed in seawater at a water depth of 1 m for 180 d, and bare epoxy resin panels and panels coated with PDMS elastomers were used as blank and reference samples, respectively. The actual marine test photographs (not cleaned) are depicted in Fig. 8a. Fewer or almost no fouling organisms were observed on all epoxy resin panels after 30 d. After 60 d, blank panels were covered with a large number of fouling organisms, but the hydrogel surface was almost free of fouling organisms. This is because the hydration layer on the surface of the hydrogel effectively prevents the adhesion of fouling organisms [72]. After 90 d, blank panels and panels coated with PDMS elastomers were completely covered with a large number of fouling organisms and only a small number of fouling organisms adhered to the hydrogel surface. Surface fouling of Gel-TA and Gel-Si-3 became progressively more severe between 120 and 180 d, but only a small number of fouling organisms were still observed on the surface of Gel-TA-Si-3. This is because the negatively charged surface, low SFE and low elastic modulus of the Gel-TA-Si-3 hydrogel synergistically prevent fouling. Additionally, this suggests that the hydrogel exhibited good resistance to biofouling within 180 d, consistent with the results of laboratory biofouling tests. Fig. 8(b-c), shows panels before and after 120 and 180 d of immersion, respectively. The results indicated that the fouling attached to the Gel-TA and Gel-Si-3 surfaces was not sufficiently cleaned, while almost no fouling organisms were attached to the Gel-TA-Si-3 surface after cleaning, indicating that it had good fouling desorption performance, which was associated with the hydration layer and low SFE of the hydrogel.

4. Conclusion

In summary, we used an eco-friendly method to prepare silicone hydrogel films with amphiphilic and negatively charged surfaces. The hydrogel retained the low surface energy properties of silicone and exhibited good mechanical and antifouling properties. The combination of a rigid portion of the polymer network (benzene rings in the PEA molecule) and a flexible part (long mPDMS chains) achieved good mechanical properties of the hydrogel. The films demonstrated good static antifouling performance in both laboratory biofouling tests and 180-day marine field tests. In addition, the covalent connection between the mPDMS monomer and the hydrogel monomer formed a robust three-dimensional crosslinked network, which improved the stability of materials in complex marine environments. This study presents new insight into the design and preparation of eco-friendly marine antifouling hydrogels. Future efforts should be devoted to the development of a multifunctional antifouling silicone hydrogel coating to fulfill the needs in actual complex marine environments.

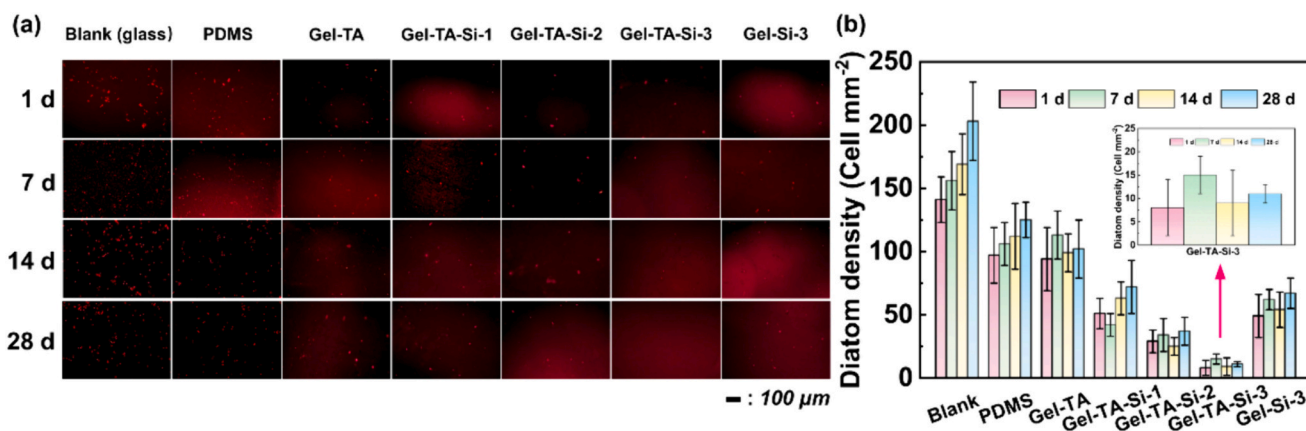


Fig. 7. (a) Fluorescence images of *Chlorella* adhered to each hydrogel surface after immersion in chlorella cell suspension for 1, 7, 14 and 21 d, the scale bar is 100 μ m. (b) The *Chlorella* density on the surface of Gel-TA-Si-x and Gel-Si-3 hydrogels.

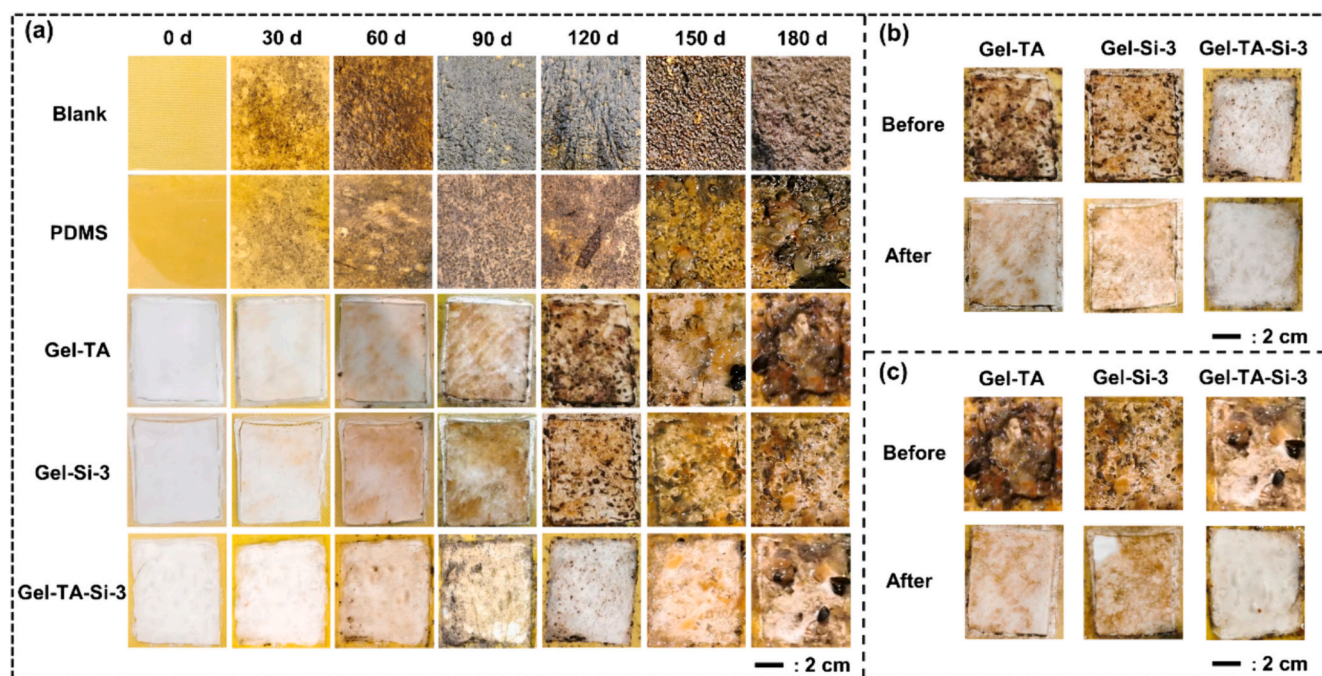


Fig. 8. Marine field test of hydrogels. (a) Images of different tested panels after 0, 30, 60, 90, 120, 150 and 180 d of immersion in seawater. (b, c) Comparison of panels before and after cleaning at (b) 120 d and (c) 180 d.

CRediT authorship contribution statement

Yuqing Zhu: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Jiawen Sun:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition. **Qingfu Wang:** Writing – review & editing, Validation, Supervision, Funding acquisition. **Jizhou Duan:** Writing – review & editing, Investigation. **Xingda Liu:** Investigation. **Ding Guo:** Investigation. **Baorong Hou:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2024.108833>.

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